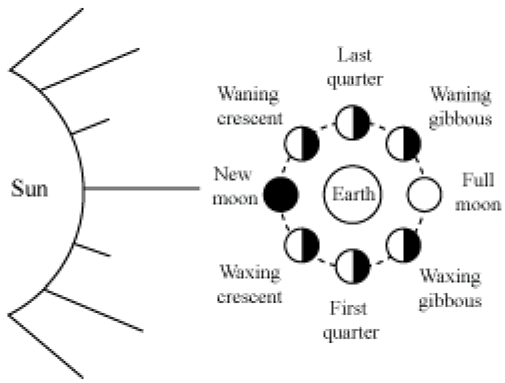
# Stars and the Solar System

- All natural objects in the sky are celestial objects.

- **The Moon**

  - New Moon day → when moon is not visible
  - Full Moon day → when full moon is visible
  - Gap between consecutive new moon day and full moon day is of 15 days.




  - Rotational period and revolution period of moon are the same (almost 29 days).

- **Moon's surface**
  - Because of lack of atmosphere, one cannot hear any sound on moon.

- **The Stars**

  - All stars emit their own light. They appear small because of large distances from the earth.

  - The sun appears bigger because it is nearer than any other stars in the space.

  - In the day time, stars are not visible because of bright sunlight.

  - Stars appear to move from east to west because of earth's rotation from west to east.

  - Pole star does not appear to move because it is very nearly situated on earth's rotational axis over the North pole.

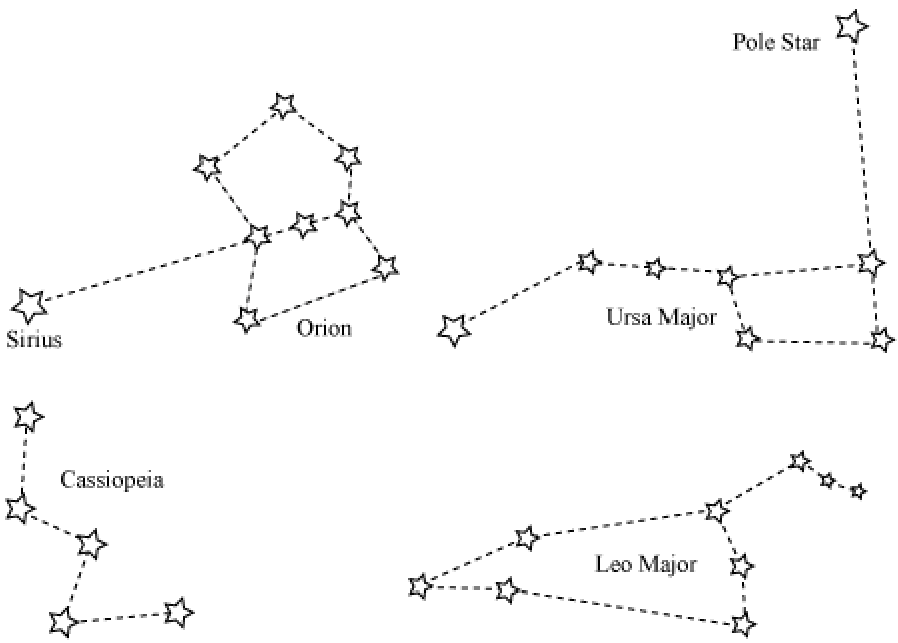
  - There are various types of stars - sun-like, red giants, super nova, twin, and variable stars.

- The imaginary line where it seems that sky and the ground are meeting with each other is known as the **horizon**.

- An imaginary sphere is formed and all the stars and planets appear moving in it. This sphere is known as **celestial sphere**.

- Celestial sphere consists of zenith, nadir, celestial poles, meridian, celestial equator and

- **Constellations**
  Groups of stars of particular shapes




- **Solar System**

- **The Sun**

  - Nearest star from the earth

- **Planets**

  - Stars twinkle in the night sky, but planets do not.
  - Planets revolve around the sun along definite paths, called orbits.
  - Time taken by a planet to complete one revolution of its orbit is called revolution period.
  - Time taken by a planet to rotate about its axis is called the period of rotation.
  - Satellites revolve around planets.

- **Inner planets**

- **Mercury**

- Nearest planet to the sun
- It is seen just before sunrise and just after sunset near horizon. It has no satellite.

- **Venus**

  - Nearest planet to the earth
  - Brightest planet in the night sky
  - Seen in the eastern sky before sunrise and in the western sky after sunset
  - Also known as morning or evening star
  - Has no satellite and rotates from east to west (sun rises in the west of Venus)

- **Earth**

  - From space, it appears blue because of 75% water content.

- **Mars**

  - It appears reddish and therefore, is known as red planet.

- **Outer planets**

- **Jupiter**

  - Largest planet in the solar system
  - Rotates very fast about its axis and has large numbers of satellites

- **Saturn**

  - Has prominent ring system and large numbers of satellites
  - Its density is less than water and is the least among the planets

- **Uranus and Neptune**

  - Both have a ring system.
  - Uranus has a tilted rotational axis and appears to roll on its side.
  - Uranus rotates from east to west similar to Venus.

- Except for mercury and venus, all the other planets have their **satellites**. Also, except for mercury, all the other planets have an **atmosphere**.

- Planets can also be categorised as terrestrial and jovian planets.

- Planets which are found inside the orbit of mars are known as **terrestrial planets.**

- Planets which are found outside the orbit of mars are known as **jovian planets.**

**Other members**

- **Asteroids**
  - Small rocky objects found in large numbers between Mars and Jupiter

- **Comets**
  - Highly elliptical objects
  - Have a bright head and long gaseous tail.
  - Tail is always directed away from the sun.
  - Halley's comet appears after every 76 years.
  - Long period comets have a longer orbital period (200 - 1000 years) while short period comets have a shorter orbital period (less than 200 years).

- **Meteors & Meteorites**
  - Objects that enter the earth's atmosphere and burn because of friction with the atmosphere
  - Large meteors that reach earth's surface are called meteorites.

- **Artificial satellite**
  - Revolves around the earth
  - Used for weather forecasting, remote sensing, communication system, etc.